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# Deep Learning for Leaf Image Sequence Alignment of Field Crops

Master's Thesis

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## Abstract

Tracking individual disease symptoms of crop leaves over time provides valuable insights into the progression of these diseases. Registering leaf images from different time points simplifies measuring the growth and emergence of individual disease symptoms over time.

We develop a landmark-based image registration method for non-rigid deformation of low-texture wheat leaves, without making use of artificial reference points. We use the feature matching method LoFTR to detect correspondences between leaf images. To validate correctness of the matches, we track matches across a three-image cycle and measure their consistency. The consistent matches then serve as landmarks for estimating a regularized thin plate spline transform that is applied to the moving image.

The resulting method does not rely on artificial markers and is thus able to process the full leaf area. Due to lack of ground truth and changes to the appearance of the leaf, quantitative evaluation of the method is challenging. Qualitatively, the method achieves satisfactory results in cases with only limited changes to the leaf, but is prone to artifacts and unstable transformations when the appearance changes drastically. The leading cause for faulty registrations are incorrect landmarks, demanding further investigation into verification of match quality.

Code is available at: <https://github.com/eboesch/LeafAlignment>

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## Chapter 1

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# Introduction

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Analyzing the progression of diseases on crops under field conditions allows the investigation of the influence of varying factors such as genotype, treatment, and environmental conditions on disease progression. Such data is challenging to obtain, however. Manual data collection is both exceedingly time consuming and prone to subjectivity. Image-based assessment methods have the potential for both objective and standardized measurements, and higher throughput and precision than manual methods. [7] However, many existing image-based assessment methods either function at canopy scale [8, 39], giving only cumulative data, or require highly standardized conditions [10, 18], making them unsuited for time series of images captured under field conditions.

Tracking individual symptoms over time provides valuable insights into the progression of diseases. One possible approach to tackle this task is to register (resp. align) all images of a time series so that they share a common coordinate system. Feature-based methods are a standard approach for image registration. They detect matching landmarks between images and estimate a transform (e.g. a homography) to align these landmarks. [30]

Like in medical imaging, organic objects such as crop leaves are often not rigid. Even with precautions such as flattening the leaf against a flat surface, non-rigid deformations still occur, e.g. through bending of the leaf or waves in the leaf's surface. Global transformations such as affine transformations or homographies are unable to correct such local deformations.

Under field conditions, the lighting conditions are subject to change between different images of the time series, making direct comparisons of image values difficult. Furthermore, the appearance of a leaf can change drastically over time: new disease symptoms appear, and existing ones grow and expand. Similarly, natural aging of the leaf ("senescence") can drastically affect the color and shape of the leaf. These changes entail both a challenge for potential feature detection between images, but also local changes in intensity, making



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comparisons between images yet more challenging. This is especially relevant with respect to evaluation of the registration.

For the specific case of wheat leaves, this already challenging task is further complicated, since they have neither distinct features or notable texture on the leaf’s surface nor a feature-rich contour that could serve as a reference for image alignment (see e.g. [24]).

Anderegg et al. [7] presented a method for solving this task relying on manually placed artificial white markers along the edges of the wheat leaves, subdividing the leaf into triangles and fitting affine transforms on the individual triangles. Their method achieved good results, however the need for the markers resulted in both a substantial amount of manual labor necessary for their placement, as well as poor registrations if markers are lost. Ensuring correct matches between the markers required time-consuming optimization of a series of parameters. [6] Furthermore, only the leaf area delimited by the markers can be registered, resulting in a loss of analyzable leaf area. For more details, see Section 2.2.

In this work, we attempt to devise a method that doesn’t rely on these artificial markers and is robust to drastic changes in appearance, thus limiting human involvement and preserving leaf area. Instead we use the local image feature matching method LoFTR to detect matches between image pairs, and use those matches as landmarks to fit a Thin-Plate-Spline transform between the images.

## Chapter 2

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# Related Work

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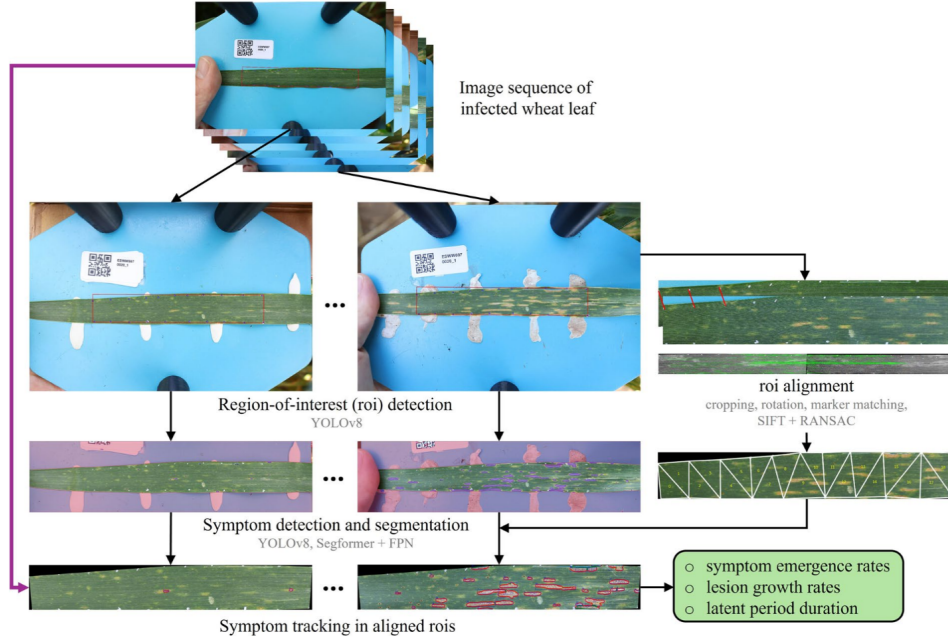
In this section we give a brief overview of image registration methods, before discussing methods used or compared to in more detailed.

### 2.1 Image Registration Methods

Feature-based methods are a standard approach for image registration. They detect sparse matching features (e.g. SIFT [26]) between images and estimate a transform (e.g. a homography) to align these landmarks. [13, 21] More recently, deep learning based methods for medical image registration make use of CNNs (VoxelMorph [9]) and transformers (TransMorph [14]) to estimate dense deformable registrations. Medical image registration faces many of the same difficulties as our use-case, such as non-rigid transformations and changes over time. However, medical images feature very complex structures, while the structure of a leaf is much simpler.

### 2.2 Piecewise Affine Approach

In this approach by Anderegg et al. [7], all images are registered to the first image of the corresponding series by using a piecewise affine transformation (see Fig. 2.1 for an overview). In a first step, white markers on the wheat leaves were detected using YOLOv8, trained on a manually annotated training set. Then a rectangular bounding box was fitted to contain all detected markers and rotated to be aligned with the image axes. This way, the region of interest (ROI) could be extracted. The markers were then matched across different frames of the time series, and the leaf sub-divided into triangles. For each triangle, an affine transformation was estimated and applied. If necessary, the image was first registered to the preceding image in the series by detecting SIFT features and using RANSAC to find the homography matrix between the images. For further details, see [6, 7].



**Figure 2.1:** Overview of image processing approach based on marker detection and piecewise affine transforms. Source: [6]

Simultaneously, the original images were individually processed using keypoint detection and segmentation models for symptom detection and segmentation, based on [40]. The resulting keypoint coordinates and segmentation masks were then aligned using the transformations obtained from the ROI-alignment. Together, this allows symptoms to be tracked based on area overlap.

The quality of the registration was assessed through manual tracking of 225 prominent features in 55 randomly selected image pairs from different time series. The average error achieved was approx. 5 pixels.

The need for artificial markers results both in a large overhead of manual workload and limits the analyzable leaf area to the ROI delimited by the markers. Furthermore, if markers (especially at the ends of the ROI) are lost to decay of the leaf or the weather, occluded, cut off due to poor placement of the leaf, or simply not detected successfully, that image segment couldn't be aligned and leaf area was lost.

## 2.3 LoFTR

LoFTR [32] is a method for local feature matching, that can produce accurate dense matches even in regions with low-texture or repetitive patterns. In contrast to classical detector-based methods (e.g. SIFT [26], ORB [31]), which first detect points of interest (e.g. corners) in both images, then extract

local descriptors of the neighborhoods of these points, and finally match the interest points based on their local neighborhoods, LoFTR doesn't use detectors. Instead, they establish pixel-wise dense matches at a coarse resolution, incorporate positional information through transformers, and finally refine the coarse matches to a finer resolution.

More specifically, they first extract both coarse-level and fine-level feature maps of both images using CNNs. The coarse feature maps are then enhanced with positional encodings and passed through a transformer module consisting of several self-attention and cross-attention layers. This allows the method to capture global context, making the transformed features context- and position-dependent. Next, a differentiable matching layer is used to match the transformed features between images, resulting in a confidence matrix. The most confident matches are selected according to a confidence threshold and mutual-nearest-neighbor criteria, yielding coarse-level match predictions. For each such prediction, a local window is cropped from the fine-level feature map and the coarse matches are refined to a sub-pixel level within this local window, resulting in the final match prediction. For more details on the method, see [32].

LoFTR stands out for its ability to find accurate dense matches even in low-texture areas. Since wheat leaves have only little texture, this makes LoFTR a prime candidate for our work.

## 2.4 Thin Plate Splines

Landmark-based thin plate spline (TPS) image registration is a classic method for non-rigid medical image registration. [19, 22, 23, 41] They were first introduced in that context by [11]. Based on an arbitrary number of corresponding landmarks  $\{p_i\}_i$  and  $\{q_i\}_i$  in the fixed and moving image, it minimizes an bending energy functional. The resulting transformation consists of a global affine component and a smooth non-linear deformation component, and is of the form

$$f(q) = P(q) + \sum_{i=1}^n w_i U(\|q - q_i\|),$$

where  $P$  is a first order polynomial,  $U$  is the radial basis function  $U(r) = r^2 \log r$ , and  $w_i$  are weights. Tahir et al. [33] use a regularized version of this minimization problem, balancing goodness of fit with smoothness of the solution, thus minimizing

$$\min_f \sum_{i=1}^n \|p_i - f(q_i)\|^2 + \lambda \iint \left[ \left( \frac{\partial^2 f}{\partial x^2} \right)^2 + 2 \left( \frac{\partial^2 f}{\partial x \partial y} \right)^2 + \left( \frac{\partial^2 f}{\partial y^2} \right)^2 \right] dx dy,$$

where the second term corresponds to the bending energy and  $\lambda$  is a hyperparameter determining the rigidity of the solution.

TPS is particularly suitable to organic shape deformations since it provides a smooth interpolation of irregularly spaced landmarks, while minimizing the bending energy penalizes high-curvature transformations resulting in physically plausible transformations. Furthermore, a closed-form solution exists for both warping and parameter estimation.

TPS is however sensitive to incorrect correspondences. Incorrect or noisy landmarks can lead to artifacts and erratic transformations. Since the radial basis function  $U$  has global support, each landmark influences the entire transformation, allowing local errors to propagate globally.

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# Materials & Methods

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### 3.1 Dataset

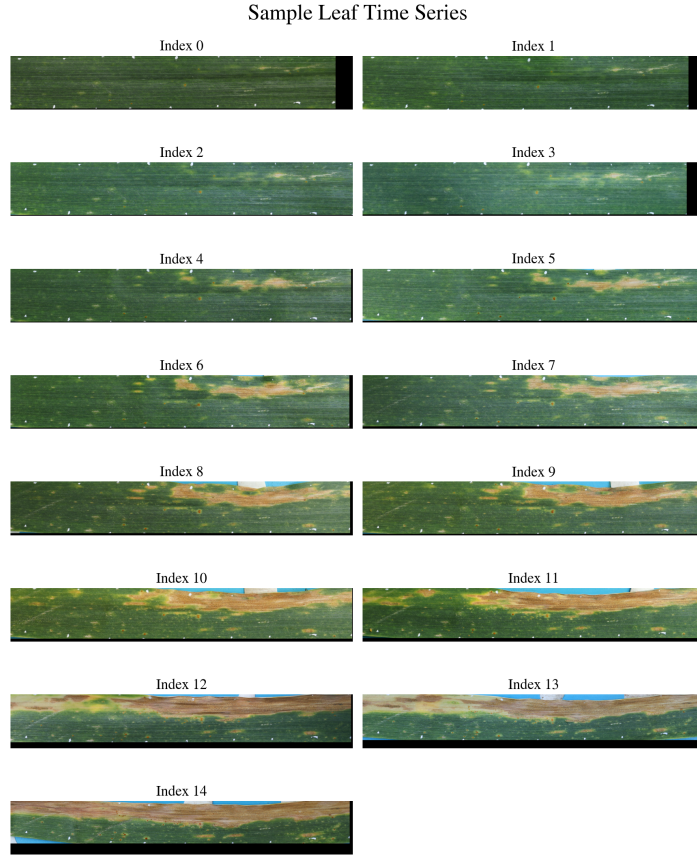
As our a dataset, we use [5], a collection of time series of high-resolution wheat leaf images with developing disease symptoms. For 1032 individual wheat leaves, photos were taken near daily directly in the field. On average, 12 images were acquired for each leaf, resulting in a total of 12'520 images. The wheat leaves were selected from different wheat cultivars with varying susceptibility to diseases, and received different treatments, resulting in a diverse dataset that captures a wide spread of characteristics. The resulting image time series capture the development and progression of foliar diseases, such as *Septoria Tritici* blotches (caused by *Zymoseptoria tritici*). [6] For an example, see Fig. 3.1.

On each leaf, 10-15 white dots were applied along both edges of the leaf to serve as reference points. In most cases these markers persist throughout the whole time series, but there are several cases where markers are lost.

For each image, the leaf is flattened against a base plate and pressed gently against poster gums to fix it in place. Together with a spacer for the camera, this ensured that the distance between camera and leaf is constant, and that the leaf and focal plane of the camera are co-planar.

The dataset provides several assets:

- the full original images,
- the same images cropped to remove irrelevant background (see Fig. 3.2a),
- segmentation masks of the cropped images, indicating background, healthy leaf and diseased areas,
- information on the Region of Interest (ROI), i.e. the area delimited by the white markers



**Figure 3.1:** Example of leaf image time series.

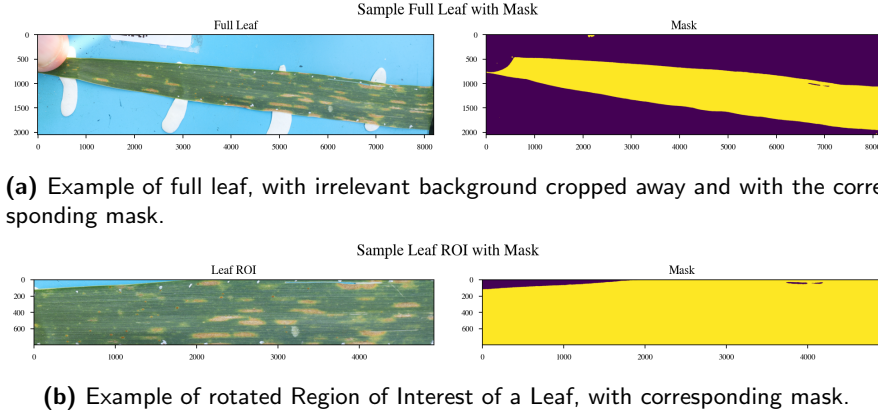
- the ROIs registered with the piecewise affine approach (see Section 2.2).
- segmentation masks of the registered ROIs.

The information on the ROIs allows us to crop the images a rectangular bounding box containing all the markers and rotate it to have an angle of  $0^\circ$ , and likewise for the segmentation masks (see Fig. 3.2b).

## 3.2 Image Pre-Processing

Very large input images can cause LoFTR to fail due to insufficient memory. We are thus forced to downscale our images to around 1.5 Megapixel. However, downscaling implies a loss of resolution, and thus also less LoFTR matches detected. To maximize the size of the leaf (while respecting hardware constraints), we crop our images to remove as much superfluous background as possible.

Using the leaf masks provided by the dataset (see Section 3.1), we mask out



**Figure 3.2:** Samples of dataset assets.

the background of each image, so that we are left with only the leaf. This ensures the method only uses information of the leaf itself, plausibly improving transferability to other datasets.

As one goal of this work is to develop a method not relying on artificial markers, we want to ensure that the markers don't play an implicit role in our method. To this end, we slightly shrink the leaf masks through erosion and then apply this new mask to the image, resulting in masked out edges. In a majority of cases, this successfully masked out the white markers.

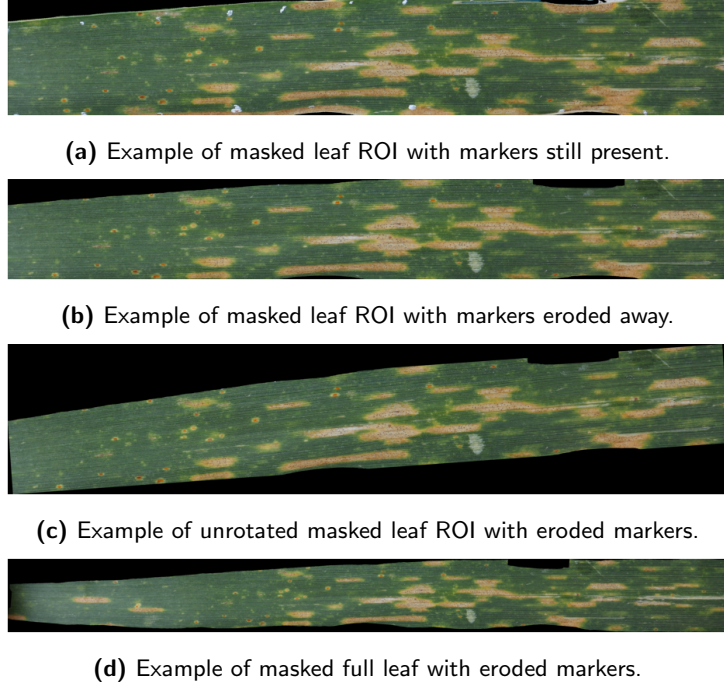
Our method supports operation on both the full leaf and the Region of Interest only. We further provide the option of "pre-rotating" the leaf images. Since wheat leaves are fairly rectangular in shape, we can fit a minimum-area bounding rectangle to the leaf mask and rotate the image so that the bounding box is aligned with the image axes. This further reduces the amount of non-leaf background, and already serves as rough baseline alignment. The method also functions without this step, however. See Fig. 3.3 for examples of the different pre-processing options.

As a final pre-processing step, all unregistered images are padded to be of the same shape. Since the TPS transform itself doesn't affect the size of the image, this is necessary for obtaining a common image size and coordinate system and allows us to compute metrics between registered images.

### 3.3 LoFTR

We use the local feature matching method LoFTR to detect correspondences between a fixed and moving image, where we want to register the moving image to the fixed image. When fitting the TPS transformation, incorrect or noisy keypoints lead to erratic transformations and poor registrations. It is thus crucial to filter out inaccurate LoFTR matches before fitting the TPS.





**Figure 3.3:** Examples of differing pre-processing options.

Inspired by [34], we devise a method we call 'Warp Consistency' that tracks matches through a three-image cycle to assert consistency of the match.

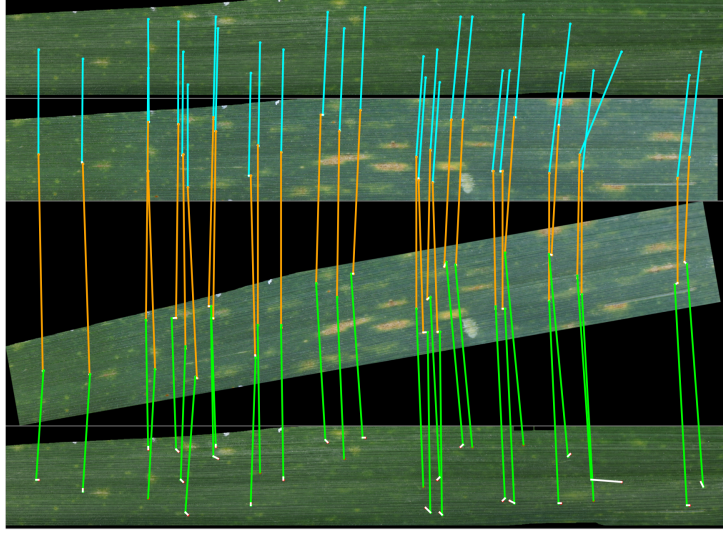
### 3.3.1 Warp Consistency

Given a fixed image  $I$  and a moving image  $J$ , we construct a third image  $J'$  by applying an arbitrary transformation to  $J$ . Next, we use LoFTR to detect matches between images  $I \rightarrow J$ ,  $J \rightarrow J'$  and  $J' \rightarrow I$ , resulting in three sets of matches

$$\begin{aligned} & \{(x_i^{I \rightarrow J}, y_i^{I \rightarrow J})\}_i \\ & \{(y_j^{J \rightarrow J'}, z_j^{J \rightarrow J'})\}_j \\ & \{(z_k^{J' \rightarrow I}, x_k^{J' \rightarrow I})\}_k \end{aligned}$$

where  $x \in I, y \in J, z \in J'$ .

For each such match  $(x_i^{I \rightarrow J}, y_i^{I \rightarrow J})$ , we search the matches  $\{y_j^{J \rightarrow J'}\}_j$  in image  $J$  from the matching  $J \rightarrow J'$  for the nearest neighbor of  $y_i^{I \rightarrow J}$ , resulting in a match  $(y_{j_i}^{J \rightarrow J'}, z_{j_i}^{J \rightarrow J'})$ . Similarly, we search the matches  $\{z_k^{J' \rightarrow I}\}_k$  in image  $J'$  from the matching  $J' \rightarrow I$  for the nearest neighbor of  $z_{j_i}^{J \rightarrow J'}$ , resulting in a match  $(z_{k_i}^{J' \rightarrow I}, x_{k_i}^{J' \rightarrow I})$ .



**Figure 3.4:** Example for cycles across three images of several matches. In order from top to bottom, the fixed image  $I$ , the moving image  $J$ , the transformed moving image  $J'$  and again the fixed image  $I$  are overlaid. Matches from  $I \rightarrow J$  are displayed in cyan, those from  $J \rightarrow J'$  in orange, and those from  $J' \rightarrow I$  in light green. Nearest neighbor matches within an image are connected with white lines. In the bottom image, the original match is shown in red and connected to its cycled counterpart with a white line. Those long white lines indicate inconsistent matches.

This yields a cycle of correspondences

$$(x_i^{I \rightarrow J}, y_i^{I \rightarrow J}) \rightarrow (y_{j_i}^{J \rightarrow J'}, z_{j_i}^{J \rightarrow J'}) \rightarrow (z_{k_i}^{J' \rightarrow I}, x_{k_i}^{J' \rightarrow I})$$

starting and ending with points  $x_i^{I \rightarrow J}, x_{k_i}^{J' \rightarrow I}$  in image  $I$  that can be compared. This chain of cycles is illustrated in Fig. 3.4.

If a match is accurate, we expect it to propagate through the images with only limited displacement, resulting in a point  $x_{k_i}^{J' \rightarrow I}$  that is close to the original point  $x_i^{I \rightarrow J}$ . In contrast, if a match is *not* accurate, we expect this displacement to be amplified as we pass through the cycle, resulting in a point  $x_{k_i}^{J' \rightarrow I}$  that is far from the original point  $x_i^{I \rightarrow J}$ . We thus compute the distance between  $x_{k_i}^{J' \rightarrow I}$  and  $x_i^{I \rightarrow J}$ , and discard all matches with distance above a threshold as inconsistent. Due to the high accuracy needed for the downstream task of tracking lesion growth, we enforce a strict tolerance of only 5px of displacement.

In [34], Truong et al. use a random transformation to create a third image. However, since we're not so much training a feature matching model, but rather evaluating it, using random transformations only introduces unnecessary uncertainty and inconsistent behavior. We thus choose to use a fixed deterministic transform.

If the transformation is too drastic, LoFTR will start struggling to find matches; we no longer test whether a match is consistent, but instead how robust LoFTR is. Transformations like homographies imply changes in scale and thus resolution, which also unnecessarily limits how many matches LoFTR can find. We thus settled on a simple rotation by  $10^\circ$ .

### 3.3.2 Orientation Correction

In some instances, a leaf was placed against the baseplate at a different orientation at one time point compared to the others during data acquisition, resulting in essentially a  $180^\circ$  rotation between fixed and moving image. For such image pairs, LoFTR will detect only few and poor quality matches, since such strong rotations lie outside its training regime. This results in completely distorted transforms and no meaningful registration.

The low number of matches found in such a case is itself a strong indicator of a  $180^\circ$  rotation between images. However, registrations between very distinct images also only produce a limited number of matches. To limit the number of the computationally expensive LoFTR runs (especially when employing Warp Consistency), we thus seek a simple and computationally cheap heuristic indicating that a  $180^\circ$  rotation between images might be present. In the case of a suspected rotation, we temporarily rotate the moving image by  $180^\circ$  and detect new LoFTR matches between the fixed image and rotated moving image. If this results in more consistent matches than with the original moving image, we claim that there is indeed a  $180^\circ$  rotation between the fixed and original moving image and proceed with the rotated moving image.

Rather than detecting the  $180^\circ$  rotation itself, we exploit that LoFTR fails in such a scenario. Since the resulting matches are of poor quality and thus geometrically inconsistent, investigating orientation consistency across triplets of matches gives a indication of LoFTR failure.

For three matches  $p_1, p_2, p_3 \in I$  and their correspondences  $q_1, q_2, q_3 \in J$ , we compute the signed area of the triangle spanned by these points:

$$A(p_1, p_2, p_3) \propto \det \begin{pmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{pmatrix},$$

where  $p_i = (x_i, y_i)$ . The sign  $A$  indicates the orientation of said triangle. [1, 29]

We then compare the orientations of the triangles in the fixed and moving images. Under a consistent geometric mapping between images, the orientation of all such triangles should behave in the same way, i.e.

$$s = \text{sign}(A(p_1, p_2, p_3)) \cdot \text{sign}(A(q_1, q_2, q_3)) \quad (3.1)$$

should have the same value for all such triplets  $(p_1, q_1), (p_2, q_2), (p_3, q_3)$ . In the case of an unstable correspondences, such as in the case of a  $180^\circ$  rotation between images, orientation will be preserved for some triangles and reversed for others.

We thus randomly sample triplets of matches and evaluate (3.1) for each sample. Triangles with negligible area are discarded to avoid numerical instabilities from nearly collinear configurations. Finally, we average over all  $s$ .

An average near zero indicates large variability in orientation of the mapped triangles, implying an unstable mapping between correspondences and thus a LoFTR failure. An average score of  $\approx 1$  suggests a stable and orientation-preserving mapping and no cause for concern. An average score of  $\approx -1$  also suggest a stable mapping, but indicates that said mapping inverts orientation, like e.g. a reflection. However, such a mapping is neither plausible for our dataset nor within LoFTR’s training regime, thus also indicating LoFTR failure.

We therefore take an average score below a threshold of 0.3 as an indication that LoFTR is failing, most likely due to a  $180^\circ$  rotations between fixed and moving image. This heuristic is computationally very inexpensive and successfully detects such rotations. While false positives do occur, they are limited in number, happen mostly in already challenging cases with large changes between fixed and moving image, and are easily dismissed by a second LoFTR call with the rotated moving image.

### 3.4 Match Filtering/Subsampling

Even after removing inconsistent matches, LoFTR often still finds well over 1000 matches, especially when matching images that are close together in time. For such large numbers of keypoints, we have experienced performance issues due to limited memory when fitting the TPS transform. We thus only use a subset of around 500 matches. We considered 4 different filtering strategies:

**Confidence** Since LoFTR provides a confidence score for each match, a straight-forward approach is to simply use the 500 most confident matches.

**Grid-wise** We divide the fixed image into grid cells and select the match with the highest confidence for each cell. Since the distribution of matches is often uneven, especially for larger gaps between fixed and moving image, not all grid cells will contain a match. We thus dynamically adjust the size (and thus number) of grid cells until we are left with roughly 500 keypoints.

**Cluster-wise** Here, we cluster the matches according to position into 500 clusters using K-Means and again pick the highest-confidence match for each cluster.

**Enforcing Minimum Distance** We sort all matches by confidence. Starting with the most confident match, we iterate through the matches and compute its distance to the previously selected matches. If the current match is too close to a previous match, it is discarded, otherwise it's added to the selected set of matches.

For all methods, we further enforce a minimum confidence require, discarding all matches with confidence lower than 0.5. If this would reduce the number of matches to less than 4, making it impossible to fit a TPS transform, we instead just use the 4 matches with the highest confidence.

### 3.5 Sequencing

Due to the at times radical changes to the leaf's appearance, both the number and quality of the matches detected by LoFTR will deteriorate the further apart the two images are. This suggests the implementation of some form of "sequencing" strategy, i.e. which preceding image a new moving image should be registered to. We consider three strategies:

**Individual** Each moving image is registered individually to the first image of the series. While this is computationally the most efficient, the quality of registrations will deteriorate as the distance between images increases, as discussed above.

**Sequential** We register the images sequentially, i.e. we estimate the registration transformation  $T_{n \rightarrow n-1}$  of  $n$ -th image  $J$  to the  $(n-1)$ -th image, and all transformations are applied sequentially:

$$\hat{J} = (T_{1 \rightarrow 0} \circ T_{2 \rightarrow 1} \circ \cdots \circ T_{n-1 \rightarrow n-2} \circ T_{n \rightarrow n-1})(J)$$

Applying a TPS transform is equivalent to a matrix multiplication. [11] As such, composing several transforms increases the computational cost linearly in the number of transformations, making this strategy computationally more expensive than individual registration. Furthermore, errors in the registration will propagate through the series, leading to more unstable registrations as more transformations are composed.

**Semi-Sequential** In order to minimize error propagation and avoid unnecessary sequencing, this strategy skips over indices, only chaining transforms when necessary. When registering the  $n$ -th image, we first attempt to register

it the the last "anchor" image and evaluate the quality of the registration based on a criterion. If the quality is deemed sufficient, we proceed on to the next image. If the registration is of poor quality, however, we set the  $(n - 1)$ -th image as the next "anchor" and register to that image instead. Finally, we concatenate the TPS transforms from one image to the next.

This strategy uses less transforms per image and thus reduces error propagation and computational cost for image warping. However, when resetting the anchor, we need to detect new matches between the moving image and the new anchor, and once again verify their quality through warp consistency. Due to the dominant computational cost of warp consistency, this strategy becomes the computationally most expensive of the three methods.

**Criterion for Registration Quality** We made several attempts of finding a criterion that could detect insufficient quality, e.g. artifacts and extreme transformations (see Section 5 for examples). We considered an approach similar to warp consistency, where we generate a third image by applying a transformation to the matches and compare the resulting warped images, but TPS is invariant under translations and rotations to either set of landmarks (see [11]) modulo interpolation differences, so this idea was discarded. We attempted a RANSAC-inspired approach, where we fit the TPS for different subsets of matches and check for inconsistencies. However due to computational costs, only few iterations could be performed, leading to unstable results unsuited for determining a threshold. Finally, while Hausdorff distance and IoU indicate extreme transformations based on mask misalignment, they cannot capture the presence of artifacts in the interior of the leaf.

Based on the observation that large regions without matches often coincide with generally low quality matches, we devise a criterion based on the distribution of matches. For each pixel, we compute the distance to the nearest keypoint. To prevent false positives due to the inherent misalignment of the existing masks, the edges of the leaf are weighed less than the interior of the leaf. Finally if the maximum distance to the nearest keypoint is above a certain threshold, coverage is considered too low and we reset anchors.

## Chapter 4

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# Experimental Setup

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All experiments were run in a [RenkuLab](#) environment with 7 CPUs, 68 GB of memory and 100 GB of storage.

All three sequencing styles were evaluated with smoothing parameters  $s = 0$  and  $s = 100$  on a test set of 100 random leaves with at least 7 data points. Only the ROI was registered, masked to erode markers, and pre-rotated, to allow for more direct comparison with the piecewise affine approach (see Sect. 2.2). Matches were filtered using warp consistency and confidence filtering. The piecewise affine method [7] was evaluated on the same test set. Furthermore, a crude baseline of comparing the plain pre-rotated ROIs was tested.

To investigate the effects of different components of the method, several variations of this default configuration with  $s = 100$  were tested on a smaller test set of 38 leaves, namely:

- not using warp consistency to validate matches
- the alternative match filtering methods (grid-based, cluster-based, minimum distance) instead of pure confidence filtering
- differing pre-processing options: not removing the markers, using the full leaf, and leaving the ROI unrotated.

The resulting registered leaf time series were inspected visually to assess quality and stability of the methods. Further, each registered image was evaluated against the first image of the sequence using a series of metrics described in the following section.

### 4.1 Evaluation

The question of how to evaluate our registrations is a challenge in its own right. [15] Since there are no ground truth registrations, direct comparison between

the predicted and ground truth registrations is not possible. In landmark-based registration, Target Registration Error is a commonly used evaluation metric. It measures the distance between the landmarks in the fixed image and the corresponding landmarks in the transformed moving image. [12, 15] However, we have no ground truth landmarks available: as established previously, the keypoints detected by LoFTR are not always accurate, even after enforcing Warp Consistency, so using them as a basis for an evaluation metric seems ill-advised. Manually identifying prominent features and tracking them (as in [7] would) yield the most meaningful metrics [15], but the amount of time necessary for that approach unfortunately goes beyond the scope of this work.

Similarly, standard segmentation-based metrics such as the Dice Similarity Coefficient [2], are not ideal for our use case, since lesions grow between time steps or new lesions appear, which these metrics don't account for.

Instead, we resort to comparing the transformed moving image directly with the fixed imaged. Due to lesions appearing and growing between time steps, this approach is inherently flawed and unable to give a definite statement of whether a registration is "good". But when evaluating different registration methods on the same data points, it should allow us to compare the different methods based on their relative scores.

We referenced a selection of image registration surveys [15, 16, 20, 28, 35] to compile a list of frequently used metrics that are applicable to our scenario, which we will introduce below. For the remainder of this section, let  $I$  be the fixed image and  $\hat{J}$  the transformed moving image.

**Mean Squared Error (MSE)** The pixel-wise mean squared error is the most straightforward loss. It's intuitive and computationally efficient. It is defined as

$$\text{MSE}(I, \hat{J}) = \frac{1}{|\Omega|} \sum_{p \in \Omega} \|I(p) - \hat{J}(p)\|^2,$$

where  $\Omega$  is the domain of valid pixels, and  $p \in \Omega$  is a pixel in the image. However, since the MSE directly compares pixel values, it is very sensitive to e.g. changes in lighting and growing lesions.

**(Patch-wise) Normalized Cross Correlation (NCC)** This metric computes the normalized cross correlation for local patches around each pixel. For two image patches  $A \subset I$  and  $B \subset \hat{J}$ , with means  $\mu_A$  resp.  $\mu_B$  and variances  $\sigma_A^2$  resp.  $\sigma_B^2$ , the local NCC is defined as

$$\text{NCC}_{\text{local}}(A, B) = \sum_p \frac{(A - \mu_A)(B - \mu_B)}{\sigma_A \sigma_B}.$$



The local NCC is then averaged over the image

$$\text{NCC}(I, \hat{J}) = \frac{1}{|\Omega|} \sum_{p \in \Omega} \text{NCC}_{\text{local}}(I_p, \hat{J}_p),$$

where  $I_p, \hat{J}_p$  are the respective patches of  $I$  and  $\hat{J}$  around pixel  $p$ . The NCC is robust to variations in brightness and contrast, making it more suitable for our scenario. [15, 16]

**Normalized Mutual Information (NMI)** Mutual information models the probabilistic relation between the pixel intensities of the images. In medical image registration, it is the primary metric used for multi-modal image registration. [15, 16] The NMI is defined as

$$\text{NMI}(I, \hat{J}) = \frac{H(I) + H(\hat{J})}{\mathcal{I}(I, \hat{J})},$$

where  $H$  is the marginal entropy

$$H(I) = - \int_{\mathbb{R}} p_I(x) \log(p_I(x)) dx,$$

and  $\mathcal{I}$  is the mutual information

$$\mathcal{I}(x, y) = - \int_{\mathbb{R}} p_{I\hat{J}}(x, y) \log(p_{I\hat{J}}(x, y)) dx dy,$$

for joint probability  $p_{I\hat{J}}$  and marginals  $p_I$  and  $p_{\hat{J}}$ . [16] The intensity distributions of both images are approximated with continuous histograms using Parzen-window estimates, see [38]. Intuitively, the NMI measures the amount of information that is shared between the two images. It is invariant to monotonic transformations of the image intensities, but it can be sensitive to noise and intensity variations. [35]

**Structural Similarity Index (SSIM)** The SSIM [37] is constructed to focus on structural similarity between images, rather than the pure pixel values. E.g. differences in brightness or contrast, while changing pixel values, preserve the perceived structure. On the other hand, e.g. blurring leads to a loss of structural information and should thus result in a lower score.

The SSIM is composed of three components: luminance, contrast and structure:

$$\text{SSIM}(I, \hat{J}) = l(I, \hat{J})^\alpha \cdot c(I, \hat{J})^\beta \cdot s(I, \hat{J})^\gamma.$$

The luminance comparison  $l$  is a function of the mean intensities of the images, while the contrast comparison  $c$  operates on the standard deviations of the images. The structure comparison  $s$  is conducted on the normalized images.

For more details, see [37]. The exponents  $\alpha, \beta, \gamma > 0$  determine the importance of the components. In practice, they are commonly set to  $\alpha = \beta = \gamma = 1$ .

All metrics introduced above are falsely improved by overlapping regions of black background. Hence, to limit the effect of differing amounts of background between leaves, we use masked versions of these metrics (with the exception of SSIM), which only take those pixels into account that are not part of the background for either image.

We further introduce two additional metrics for measuring the alignment of the image masks. These metrics of course don't capture internal misalignments, but poor scores are a strong indicator for a poor registration.

**Intersection over Union (IoU)** Also known as the Jaccard index, this metric is a normalized measure of overlap between two sets. [4] Denoting by  $M_I, M_{\hat{J}}$  the respective masks of image  $I$  and  $\hat{J}$ , it is defined as

$$\text{IoU}(I, \hat{J}) = \frac{|M_I \cap M_{\hat{J}}|}{|M_I \cup M_{\hat{J}}|}.$$

**Hausdorff Distance** The Hausdorff distance measures how far two sets are from each other. It is defined as

$$\text{Hausdorff}(I, \hat{J}) = \max \left\{ \sup_{p \in M_I} d(p, M_{\hat{J}}), \sup_{q \in M_{\hat{J}}} d(q, M_I) \right\},$$

where  $d(q, M_I) = \inf_{p \in M_I} d(p, q)$  is the distance of point  $q$  to the set  $M_I$ . [3] Intuitively, it measures the "worst case" misalignment of the two masks. In practice, one usually uses the 95th percentile of the Hausdorff distance to make the metric more robust to outliers.

Note that, since there are variations in the fitting of the masks to the leaf, these metrics can at times be misleading. For example, when registering the full leaf (rather than just the ROI), the different images of a time series will not always contain the exact same area of the leaf. In one image, the leaf might be placed further to the right, cutting off leaf area that was present in the preceding image. This will lead to misalignment of the masks, even if the registration of the shared leaf area is perfect.

## Chapter 5

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# Results

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Our registration method and the piecewise affine method frequently fail on the same cases, primarily those with drastic changes of appearance. However, both methods can succeed in cases where the other fails. Our method becomes unstable for many leaves, predominantly at the later stages, developing artifacts, instability and large deformations. Some examples of such flaws can be seen in Fig. 5.1. The piecewise affine method generally produces very stable registrations with no artifacts. It does, however, frequently lose leaf area due to lost markers, especially at the corners of the ROI. In the most extreme cases, the entire leaf is lost. This leads to poor IoU and Hausdorff distance, though even good registrations show a baseline mask misaligned because the masks of the first image in the sequence appear to be less tightly fitted to the markers than for subsequent images.

All methods achieve worse scores for higher indices, see Fig. 5.3, which might indicate a loss of performance. However, since higher indices commonly coincide with larger changes to the leaf, all metrics but IoU and Hausdorff distance would deteriorate naturally even for a perfect registration.

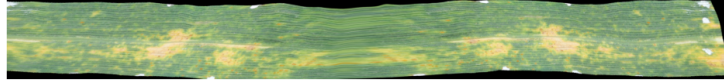
**Including markers yields only slight increase in robustness.** Including the markers rather than removing them via masking appears to slightly increase stability, being somewhat less prone to artifacts and tends to preserve leaf area better. However, the same issues of artifacts, small local deformations and over-exaggerated transforms persist. When not pre-rotating the ROI the quality remains similar in a majority of cases. Failure cases tend to be more extreme, however. When registering the full available leaf, results within the ROI are predominantly consistent with those of ROI-only registration. Beyond the ROI, the transformations tends to be more unstable, likely due to difference in leaf area present in the original image. These differences also lead to comparatively poor IoU and Hausdorff distance, see Tab. 5.2. The resulting registrations can at times be more stable than ROI only.



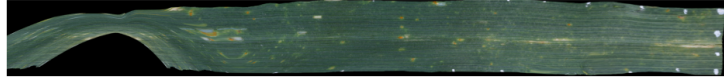
(a) Unstable transformation, indicated by the fluctuating center line



(b) Unstable transformation causing a circular artifact about a third off the left edge.



(c) Unstable transformation leading to artifacts that seem to almost mirror the leaf.



(d) Unstable registration leading to extreme deformations on the left-hand side of the leaf.

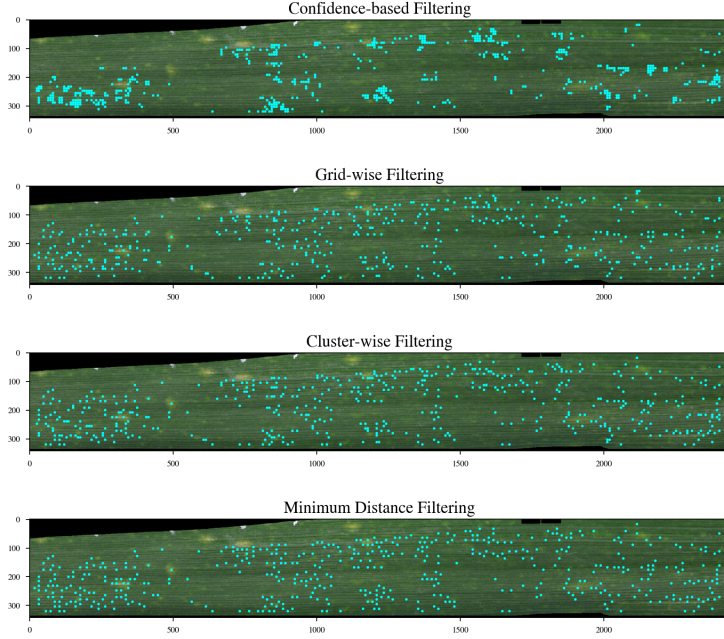
**Figure 5.1:** Examples artifacts and unstable transformations produced by our method.

**Warp Consistency greatly improves stability of the method.** As every LoFTR call takes 3-5s, Warp Consistency is the computationally most expensive step of our method. We found that, for a tolerance threshold of 5px, on average  $\approx 25\%$  of matches pass this warp consistency test, with higher percentages in easier cases and lower percentages in more challenging cases. From a visual inspection of these points (henceforth referred to as *consistent*), we find that the amount of clear mismatches is drastically reduced but not all incorrect matches are caught.

When not ensuring warp consistency of the matches, the registrations are exceedingly unstable, with artifacts appearing in the vast majority of test cases. Though for leaves with limited changes in appearance, results similar to those utilizing warp consistency may be achieved. Unfortunately, this instability is not at all reflected in the metrics (see Tab. 5.2), as it consistently outperforms the default configuration with warp consistency. This puts into question the legitimacy of our metrics.

Using a random homography transformation rather than a fixed  $10^\circ$  rotation during warp consistency to similar quality, though more inconsistent, likely due to randomness of the transformation.

**Filtering styles lead to no significant difference.** Due to LoFTRs coarse-to-fine approach of detecting matches, the confidence of a match is correlated to that of its neighbors, meaning that high-confidence matches most often appear



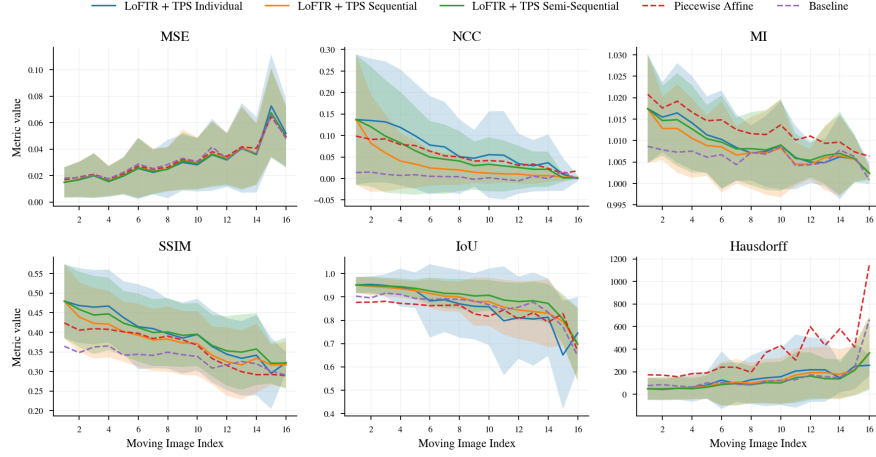
**Figure 5.2:** Example of match distributions for different filtering styles.

in clusters. Confidence-based filtering thus results in very uneven distributions of keypoints across the leaf. Both grid- and cluster-wise filtering result in a more even distribution of keypoints than confidence-based, but keypoints can still cluster together at edges and corners of the grid cells. Minimum distance filtering achieves the most even distribution of landmarks, as the enforced minimum distance prevents any clustering. Example distributions of the different filtering styles can be seen in Fig. 5.2

The differing filtering styles resulted in predominantly very similar results. In some cases, the more even distributions of grid-wise, cluster-wise or minimum distance filtering can help with maintaining the general shape of the leaf, but they tend to be more prone to artifacts and small local deformations.

**Regularization improves stability drastically.** Regularization with a smoothing parameter of  $s = 100$  greatly reduces the amount of small local deformations and thus improves stability of the method compared to no regularization. Indeed, it consistently outperforms unregularized registrations across all metrics and sequencing styles, see Tab. 5.1.

**Semi-Sequential registration shows most potential.** Of the three sequencing styles, semi-sequential registration produces the most stable transformations, as indicated by superior performance in IoU and Hausdorff distance, see Fig. 5.3, Tab. 5.1. Drastic changes to the leaf (i.e. for higher indices) still produce



**Figure 5.3:** Average performance over different time steps for the different registration methods. Each subplot shows a different metric. For MSE and Hausdorff distance, lower values correspond to better performance; for the other metrics higher values are better.

artifacts and exaggerated transforms, leading to a decay in performance across metrics for higher indices. However, these artifacts are less common and less severe than for sequential and individual registration. In many cases, sequential registration produces very similar results to semi-sequential registration, likely to do the the anchor being reset frequently during semi-sequential registration. Small local deformations are however more common and more pronounced due to increased error propagation, leading to poorer scores than semi-sequential registration. Accordingly, the registration can become unstable earlier in the series. The lack of error propagation in individual registration leads to better results in select cases, especially in the earlier stages. Indeed, for lower indices individual registration outperforms semi-sequential registration for most metrics. However, individual registration is generally far more prone to artifacts than the other sequencing styles, leading to poor IoU and Hausdorff distance. The instability is further reflected in the higher variance of the metrics. In many cases, the last images in the sequence become completely unusable.

**Table 5.1:** Average Scores for the different registration methods and different smoothing parameters.

| Registration Method | Smoothing | MSE           | NCC           | MI            | SSIM          | IoU           | Hausdorff    |
|---------------------|-----------|---------------|---------------|---------------|---------------|---------------|--------------|
| Baseline            | -         | 0.0264        | 0.0062        | 1.0068        | 0.3451        | 0.8871        | 102.75       |
| Piecewise Affine    | -         | 0.0258        | 0.0645        | <b>1.0146</b> | 0.3850        | 0.8565        | 264.00       |
| Individual          | 0         | 0.0238        | 0.0650        | 1.0099        | 0.3990        | 0.8897        | 106.71       |
|                     | 100       | <b>0.0238</b> | <b>0.0874</b> | 1.0110        | <b>0.4217</b> | 0.8949        | 106.67       |
| Sequential          | 0         | 0.0254        | 0.0190        | 1.0070        | 0.3108        | 0.8110        | 129.57       |
|                     | 100       | 0.0246        | 0.0414        | 1.0095        | 0.3972        | 0.9079        | 96.13        |
| Semi-Sequential     | 0         | 0.0250        | 0.0409        | 1.0085        | 0.3508        | 0.8555        | 108.33       |
|                     | 100       | 0.0242        | 0.0655        | 1.0107        | 0.4165        | <b>0.9221</b> | <b>84.22</b> |

**Table 5.2:** Average Scores for varying experimental configurations of semi-sequential registration.

| Configuration                 | MSE    | NCC    | MI     | SSIM   | IoU    | Hausdorff |
|-------------------------------|--------|--------|--------|--------|--------|-----------|
| Default ( $s = 100$ )         | 0.0230 | 0.0766 | 1.0105 | 0.4131 | 0.9251 | 77.22     |
| No Consistency                | 0.0228 | 0.0980 | 1.0112 | 0.4199 | 0.9341 | 56.31     |
| Min. Dist. Filter             | 0.0228 | 0.1067 | 1.0117 | 0.4285 | 0.9319 | 65.42     |
| Cluster Filter                | 0.0228 | 0.1047 | 1.0116 | 0.4270 | 0.9319 | 64.95     |
| Grid Filter                   | 0.0228 | 0.1066 | 1.0116 | 0.4290 | 0.9316 | 65.13     |
| With Markers                  | 0.0250 | 0.0606 | 1.0135 | 0.4283 | 0.9392 | 66.90     |
| Full Leaf                     | 0.0250 | 0.0433 | 1.0090 | 0.4618 | 0.8721 | 142.84    |
| Unrotated ROI                 | 0.0225 | 0.0707 | 1.0108 | 0.6450 | 0.9075 | 66.23     |
| Unrotated ROI<br>(Homography) | 0.0229 | 0.0615 | 1.0104 | 0.6344 | 0.8972 | 70.03     |

## Chapter 6

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# Discussion

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As evidenced by the drop in stability when not using warp consistency, it is clear that the quality of the matches is crucial to the quality of the registration. While warp consistency catches a majority of outliers, further criteria for evaluating the quality of matches would greatly be beneficial.

Alternatively, the quality of LoFTR matches could be addressed directly. EfficientLoFTR [36] builds on LoFTR, achieving faster inference and a slight increase in accuracy while preserving LoFTR’s strengths of detecting matches even in low-texture areas. While not guaranteed to find higher quality matches, this would make repeated iterations of Warp Consistency possible, making it more robust.

The combination of the feature extractor SuperPoint [17] and the feature matcher [25] is another popular LoFTR alternative, due to its efficiency and robustness. [27] However, this approach can struggle if the subject has low texture, like in our case, making it a less popular candidate.

The low texture of wheat leaves seems like a plausible cause for low quality matches. However, in cases with little to no changes to the leaf’s appearance our method seems to consistently perform well (though the lack of texture makes it difficult to assess well), making this hypothesis more unlikely.

The masking of the leaves seems to introduce some uncertainty of its own. The fluctuation in leaf area distort our mask based metrics even for high quality registrations. And since the edges of a mask aren’t always fitted to the markers in the same way, even qualitative visual assessment are non-trivial. Furthermore, the eroding the mask to remove markers also expands gaps in the masks, leading to holes in the masks.

Regarding the choice of transformation for warp consistency, since we use only a single iteration, choosing a deterministic transformations is preferable, since it leads to more consistent results. As such, pre-rotating the ROI is also



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worthwhile, since it allows the rotation during warp consistency to introduce a consistent degree of change, rather than one subject to the leaf’s existing orientation.

Note that filtering by distribution rather than confidence results in lower confidence matches being used compared to filtering by confidence. Since confidence can be considered a proxy for the quality of the matches, lower match quality likely also explains the slight drop in stability when filtering by distribution. It stands to reason that when using less landmarks for TPS, their distribution becomes more important, in order to ensure the whole leaf is affected correctly. For roughly 500 matches even filtering by confidence results in matches covering the majority of the leaf.

Quantitative evaluation of the method is another main concern. None of the metrics used seem to reliably capture the observed instabilities and artifacts of our method, and at times directly contradict our qualitative observations, e.g. in the case of no warp consistency. This makes it very challenging to objectively determine which method is better, and which components have how much influence.

A metric making use of the full segmentation mask could plausibly detect artifacts that tear apart lesions. Such a metric would, however, have to take the growth and appearance of new lesions into account. For a definitive assessment of the quality of registration comparable to [7], a manual evaluation is likely necessary.

Additionally to evaluation, a method to detect unstable registrations would be of great value, as it could also be used as a criterion for when to reset anchors during semi-sequential registration. The current criterion based on distribution of matches is only a heuristic based on observations, rather than a true measure of registration or transformation quality. Furthermore, it is often times too quick to reset anchors, leading to unnecessary error propagation. Given the highly volatile transformations, one possible approach would be some sort of smoothness resp. regularity measure, such as TPS’s bending energy or the determinant of the Jacobian. [15]

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# Conclusion

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We constructed a method for image registration of wheat leaf images using LoFTR and thin plate splines that doesn't rely on artificial markers. To ensure the quality of matches detected with LoFTR are of sufficient quality, we devised a method we call Warp Consistency that tracks matches across a three image cycle to measure their stability. Warp Consistency improves stability of the registrations significantly. Our method successfully registers series of leaf images without relying on artificial markers, provided the changes in the leaf are not too severe. For more drastic differences in appearance, the method suffers from instability and artifacts, primarily due to poor quality of LoFTR matches. Further methods to validate the quality of matches should greatly benefit our method.

Quantitative evaluation of the method is challenging, as there is not ground truth to compare to. Direct comparisons between fixed and registered moving images are only of limited use, since a majority of metrics cannot account for growing and appearing changes to the leaf. Furthermore, none of the considered metrics or methods could reliably detect artifacts caused by poor TPS transformations. A method to quantitatively assess the quality of the registration and/or detect artifacts and unstable transformations is crucial for further improvement of our method.

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## AI Usage Card for Project



### CORRESPONDENCE(S)

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ETH Zürich

### PROJECT NAME

Deep Learning for Leaf Image Sequence Alignment of Field Crops

### KEY APPLICATION(S)

The tasks and applications the project.

### LITERATURE REVIEW

ChatGPT, GPT-5

### USAGE CONTEXT

Finding references for related work and supporting literature.

### CONTROL

All references were verified w.r.t. content and applicability. Only relevant references were used.

### CODING

ChatGPT, GPT-5

### USAGE CONTEXT

Code generation for minor utility tasks and data visualizations (matplotlib) based on concrete instructions.

### CONTROL

Generated code was tested, and independently corrected and adjusted.

THE AUTHORS VERIFY AND AGREE WITH THE MODIFICATIONS OR GENERATIONS OF THEIR USED AI-GENERATED CONTENT





Eidgenössische Technische Hochschule Zürich  
Swiss Federal Institute of Technology Zurich

## Declaration of originality

The signed declaration of originality is a component of every written paper or thesis authored during the course of studies. **In consultation with the supervisor**, one of the following two options must be selected:

- ☐ I hereby declare that I authored the work in question independently, i.e. that no one helped me to author it. Suggestions from the supervisor regarding language and content are excepted. I used no generative artificial intelligence technologies<sup>1</sup>.
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### Title of paper or thesis:

Deep Learning for Leaf Image Sequence Alignment of Field Crops

### Authored by:

*If the work was compiled in a group, the names of all authors are required.*

#### Last name(s):

Bösch

#### First name(s):

Elena

With my signature I confirm the following:

- I have adhered to the rules set out in the [Citation Guidelines](#).
- I have documented all methods, data and processes truthfully and fully.
- I have mentioned all persons who were significant facilitators of the work.

I am aware that the work may be screened electronically for originality.

#### Place, date

13/05/2026

#### Signature(s)

Elena Bösch

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<sup>1</sup> For further information please consult the ETH Zurich websites, e.g. <https://ethz.ch/en/the-eth-zurich/education/ai-in-education.html> and <https://library.ethz.ch/en/researching-and-publishing/scientific-writing-at-eth-zurich.html> (subject to change).